



Cost of Living in the US

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Overview

- Around 31,000 observations
- 15 variables
- Dropped null values
 - Only 10 observations lost
- Tools used: Python with the packages Pandas, Seaborn, Plotly, Matplotlib, Folium, Sklearn

Variables

Categorical

- caseID
- state
- isMetro
- areaname
- county
- family_member_count

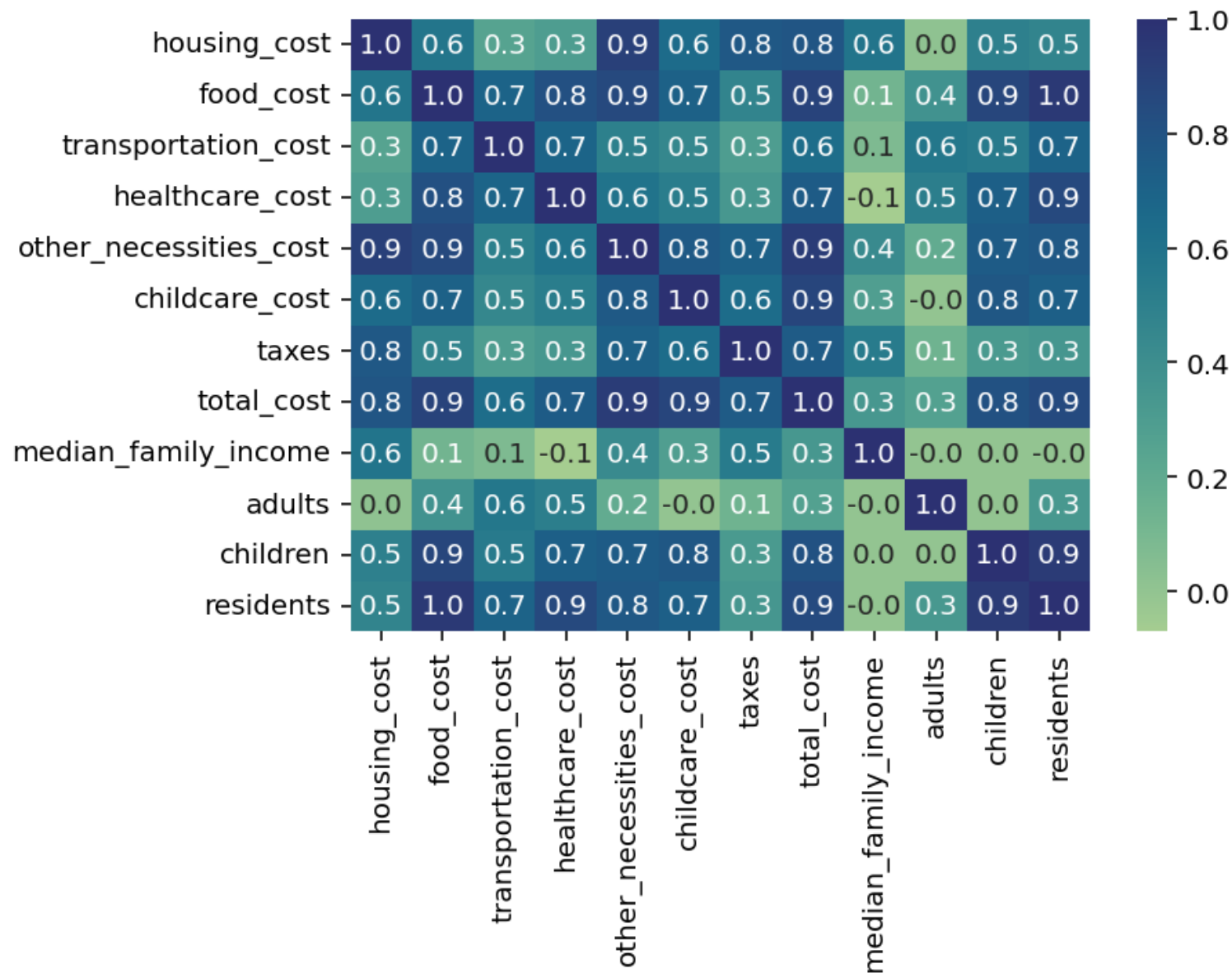
Quantitative

- childcare_cost
- housing_cost
- food_cost
- transportation_cost
- healthcare_cost
- other_necessities_cost
- taxes
- total_cost
- median_family_income

Family Member Count

- Formatted like “1p3c”
- Variables created
 - adults – parent number
 - children – children number
 - residents – parents + children

Correlation Heatmap



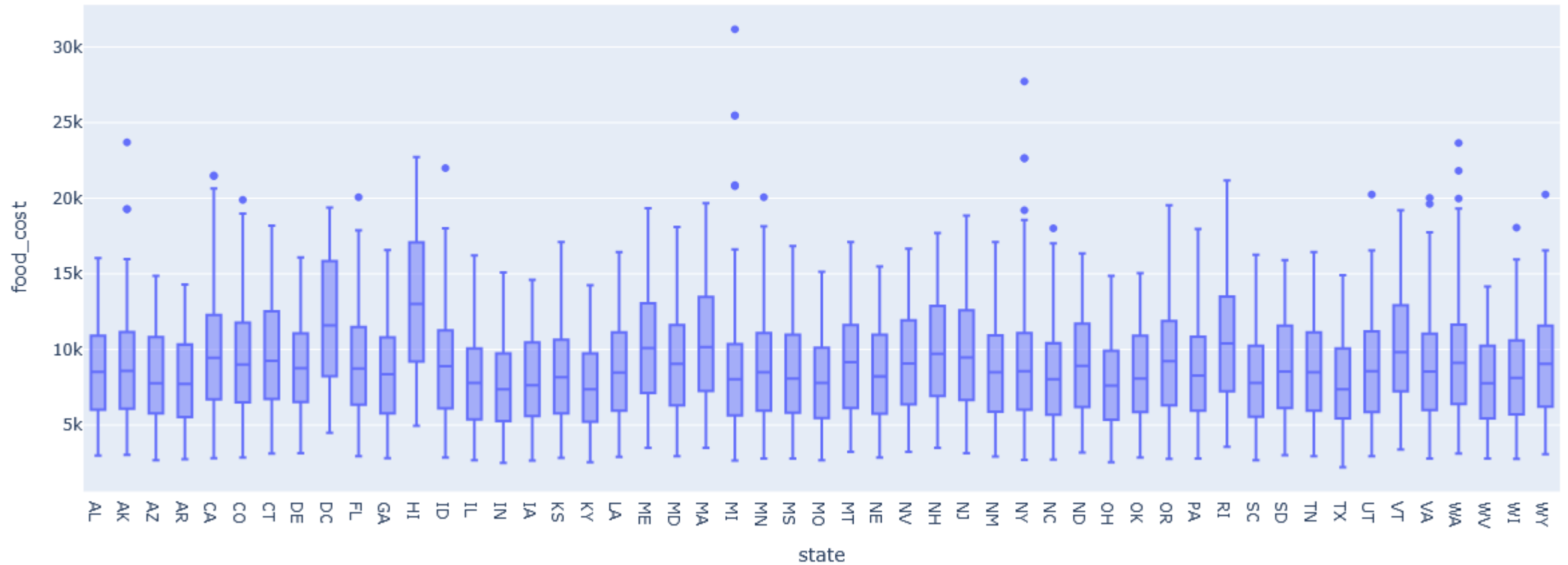
Top 12 Taxes & Childcare Cost by County

county	taxes
San Mateo County	34738.964880
Marin County	31637.529360
San Francisco County	27174.135960
New York County	26517.054360
Santa Clara County	24651.857760
Rockland County	23463.096600
Queens County	22807.598208
Norfolk County	22347.509448
Westchester County	22292.750676
Suffolk County	22256.451666
District of Columbia	21744.785220
Honolulu County	21423.277428

county	childcare_cost
District of Columbia	31940.492400
New York County	29620.246800
Queens County	28050.334320
Rockland County	27260.377200
Westchester County	26661.030720
Suffolk County	25288.633680
Bronx County	24340.542960
Norfolk County	23606.173680
Dutchess County	22844.859600
Ulster County	21947.080944
Tompkins County	21589.945152
Grand Forks County	21525.153984

Food Cost by State

Food Cost by State



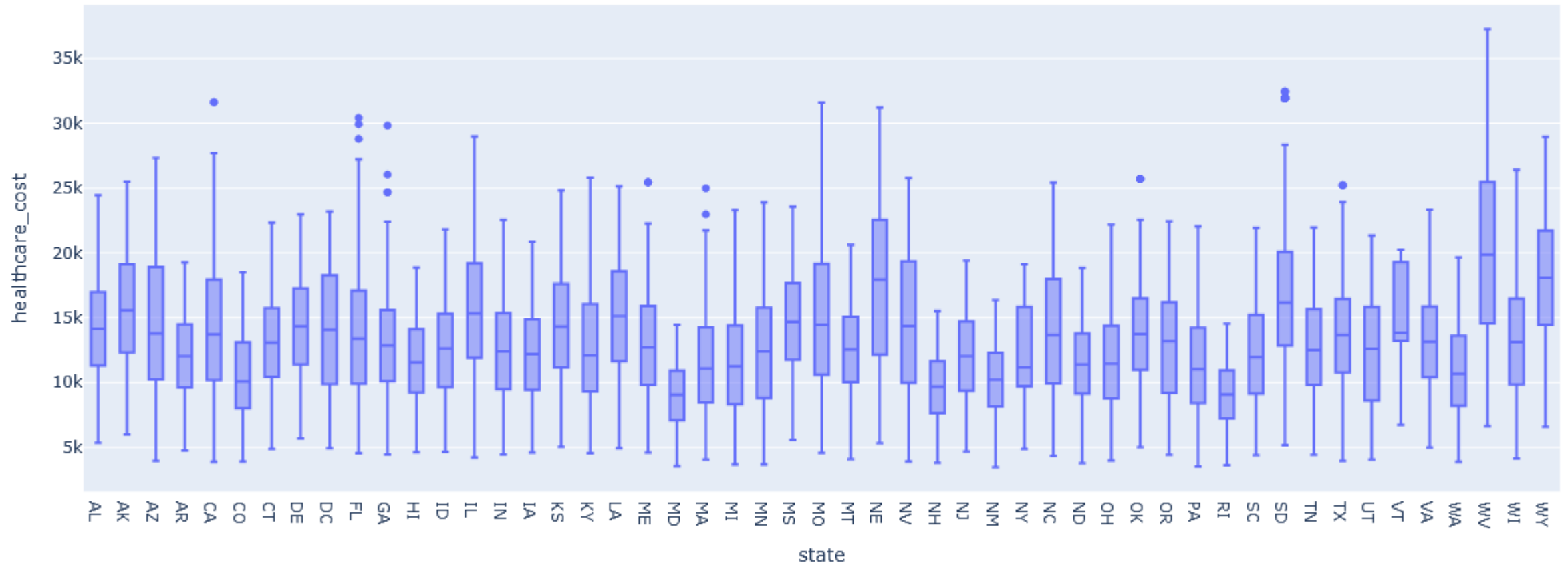
Transportation Cost by State

Transportation Cost by State

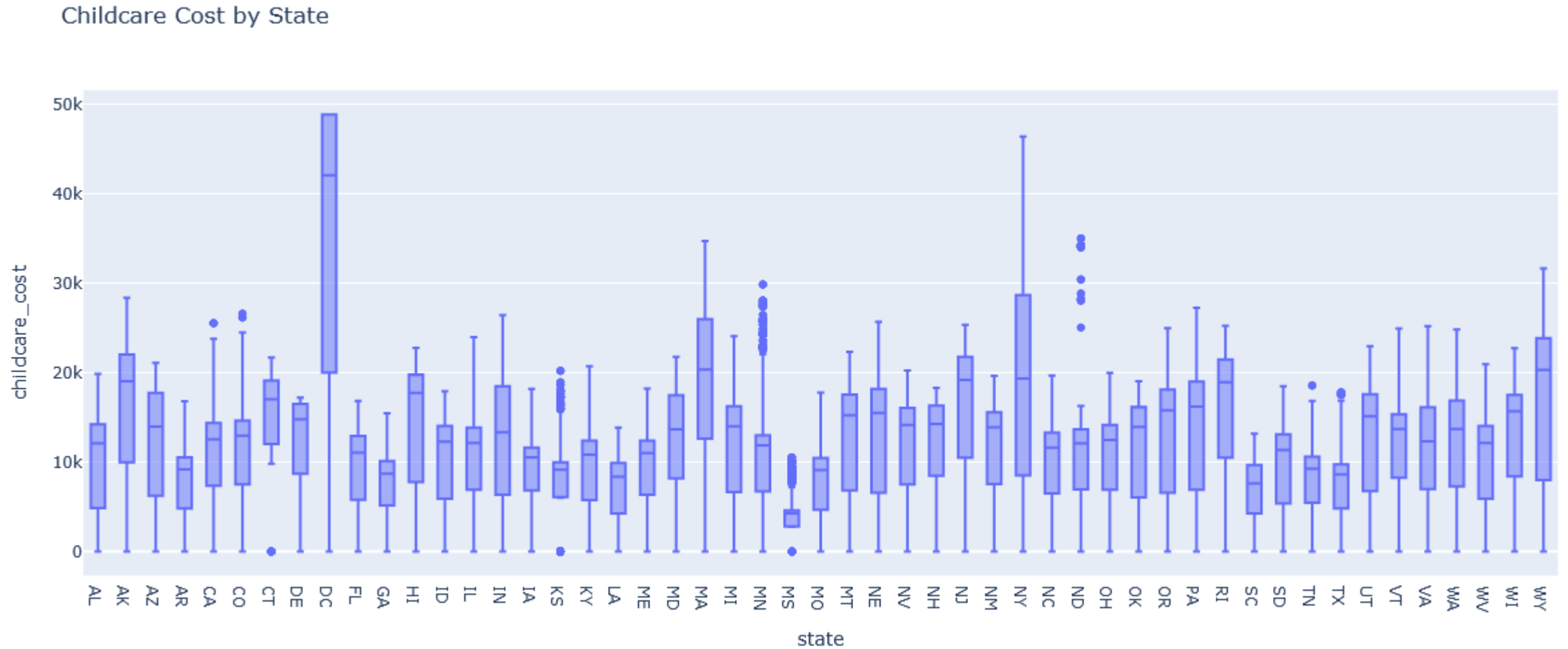


Healthcare Cost by State

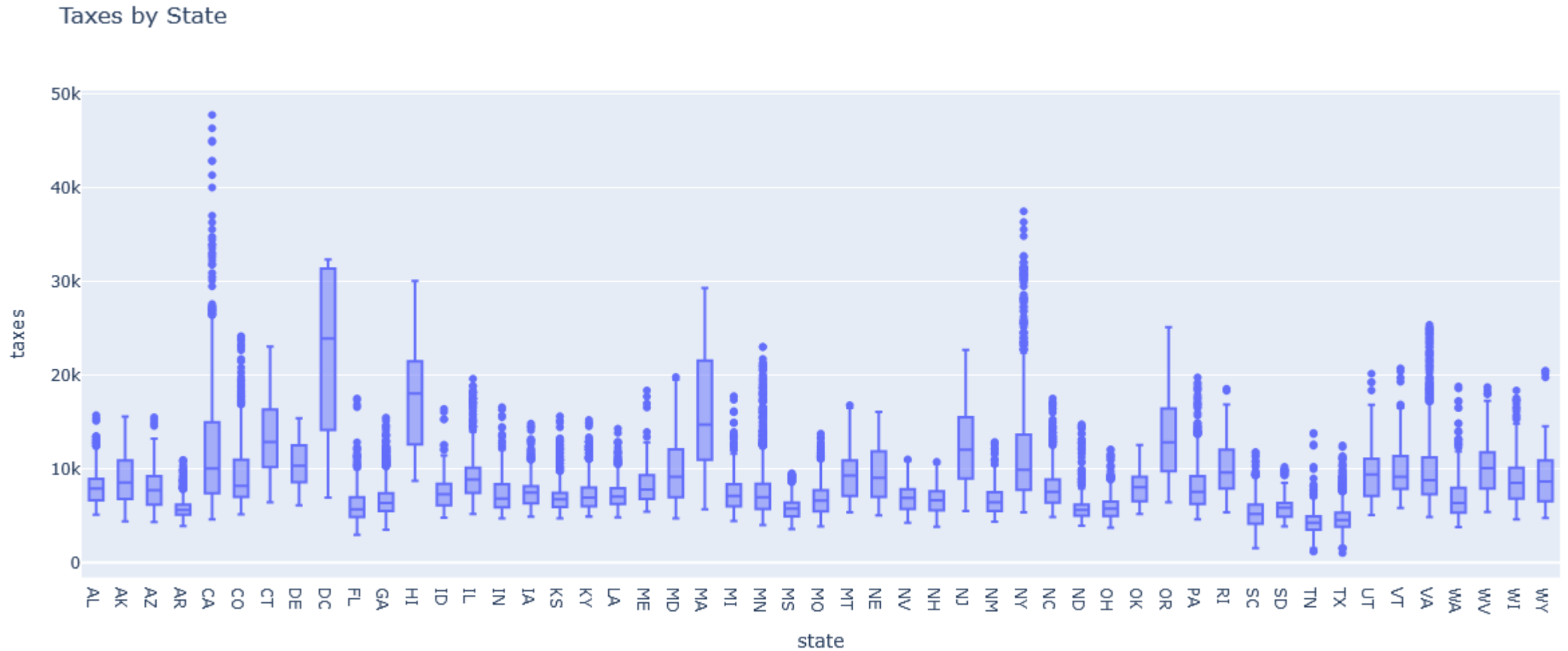
Healthcare Cost by State



Childcare Cost by State

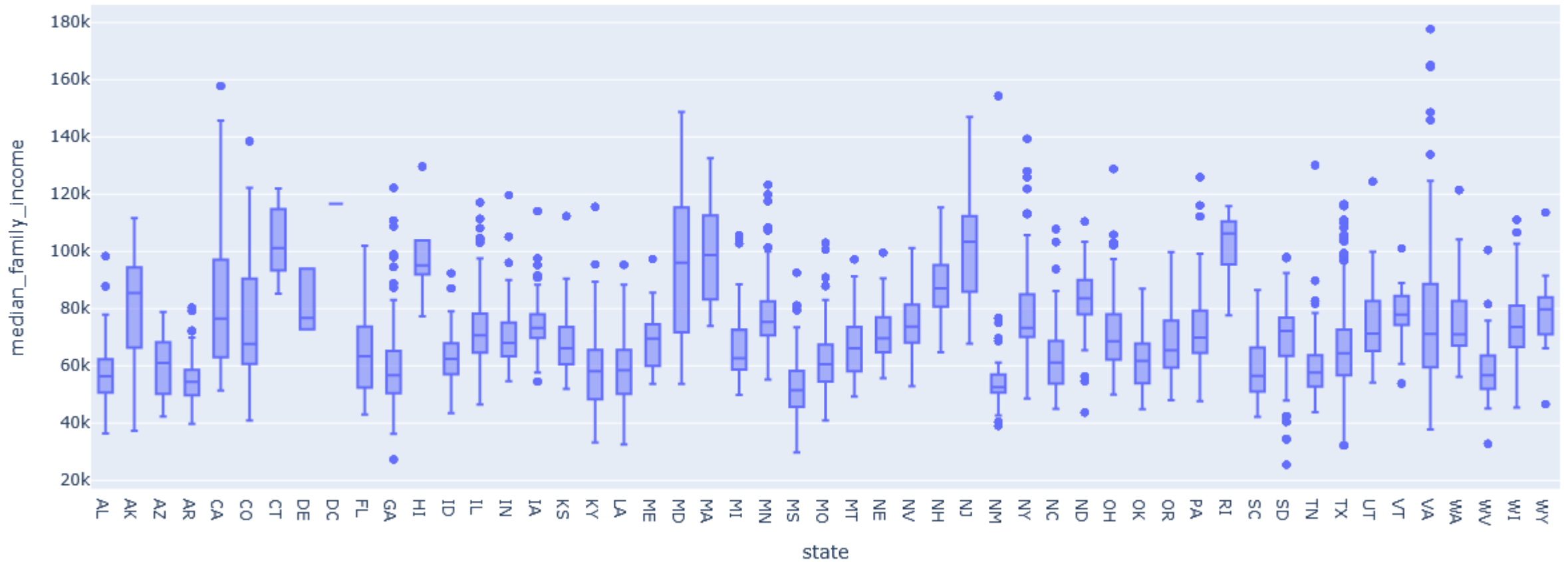


Taxes by State



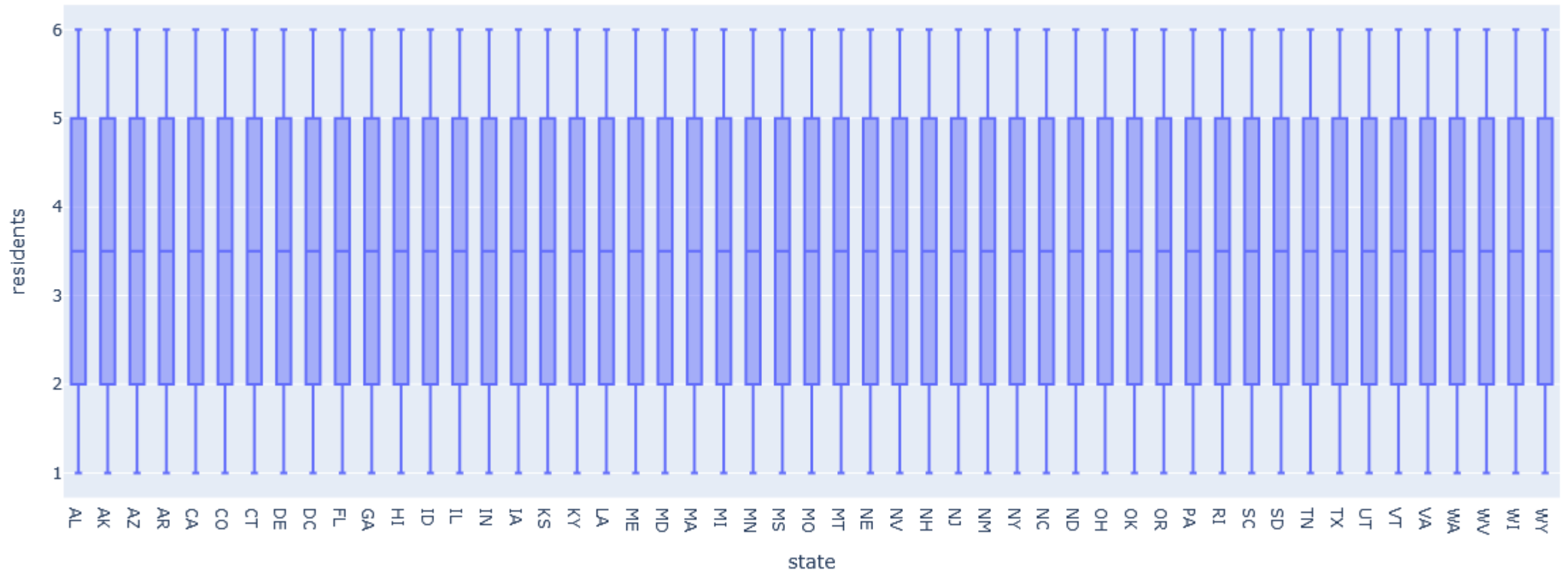
Median Family Income by State

Median Family Income by State



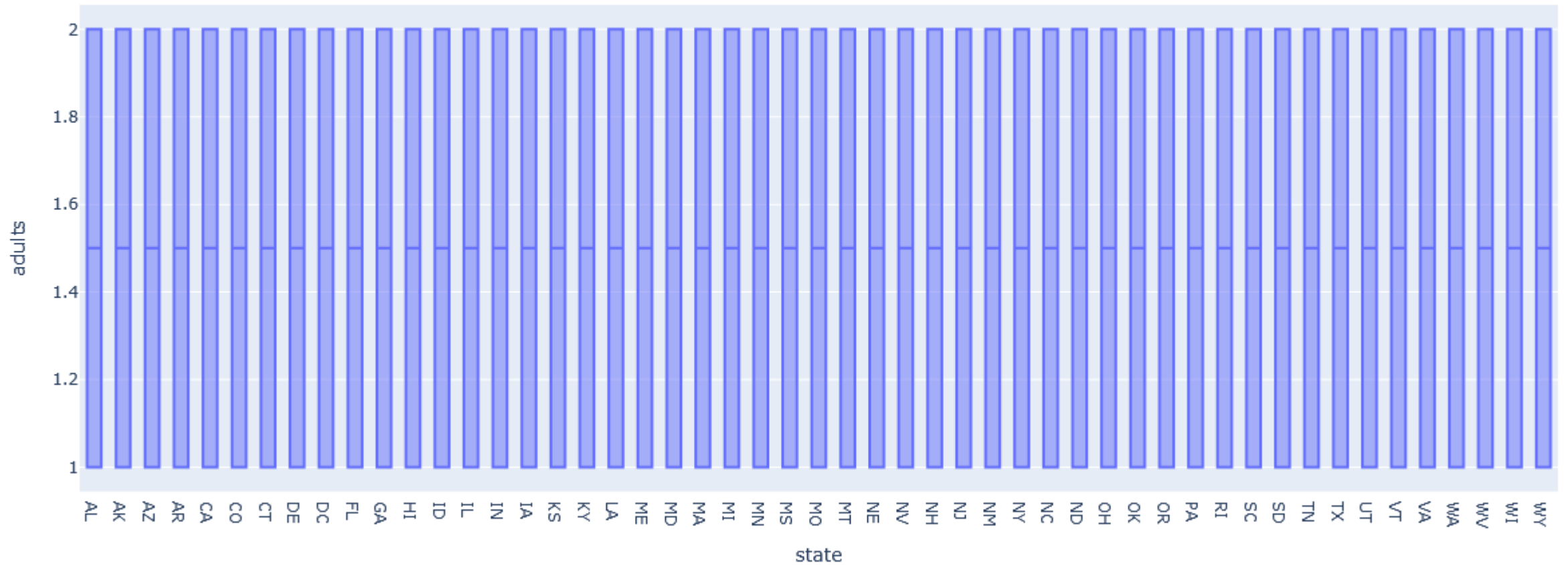
Residents by State

Total Residents by State



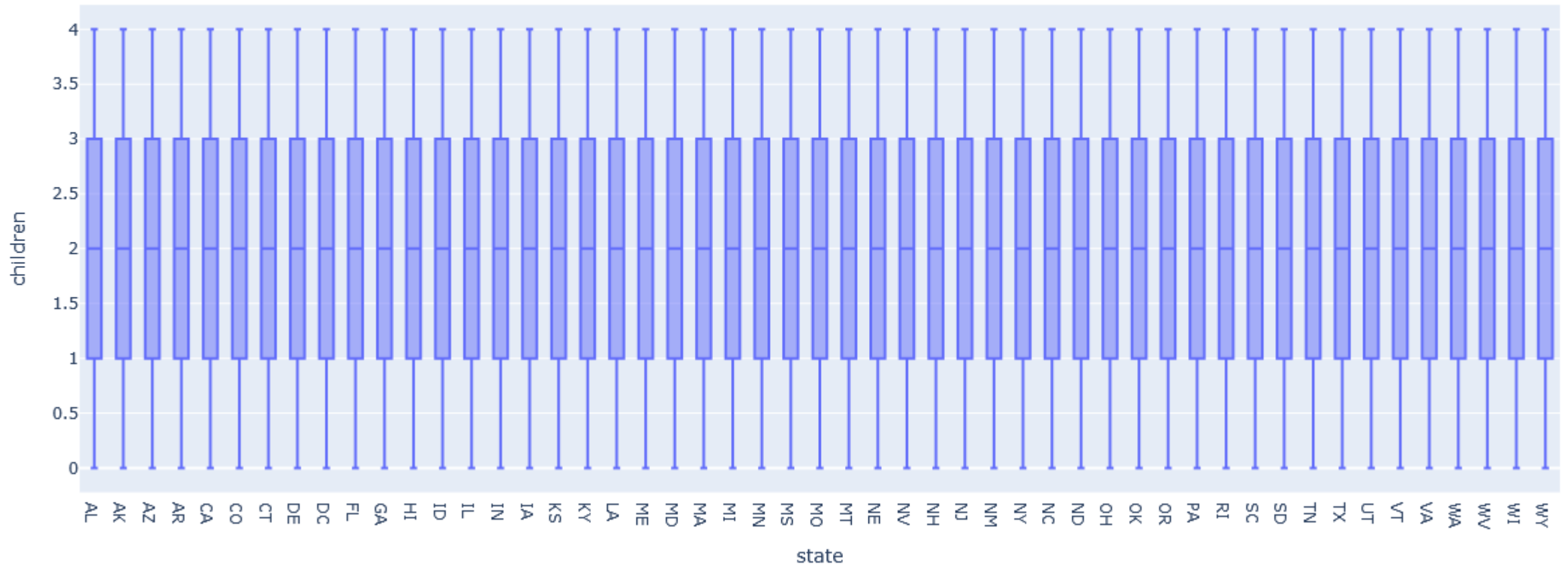
Adults by State

Adults by State

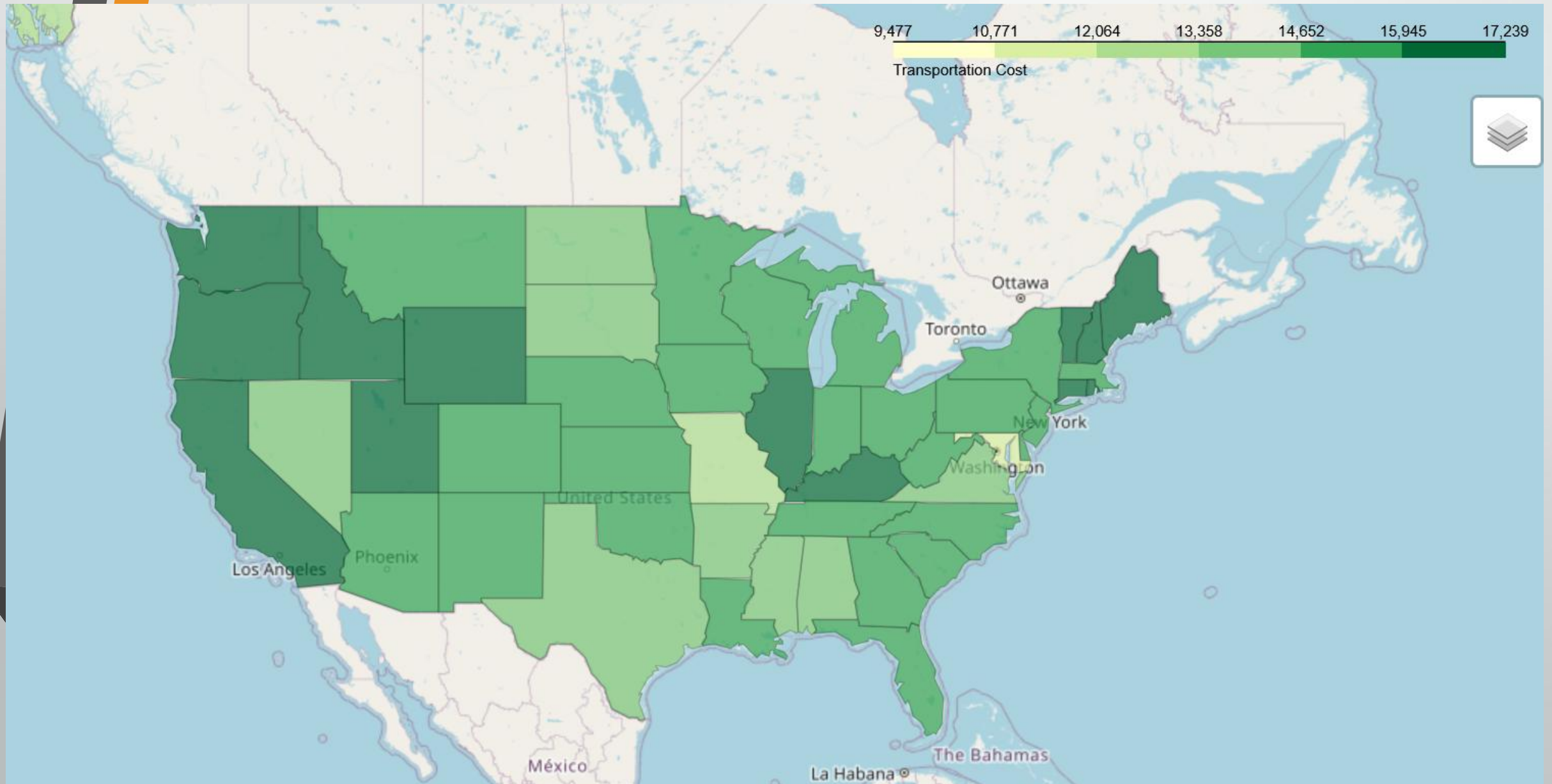


Children by State

Children by State



Transportation Cost by State map



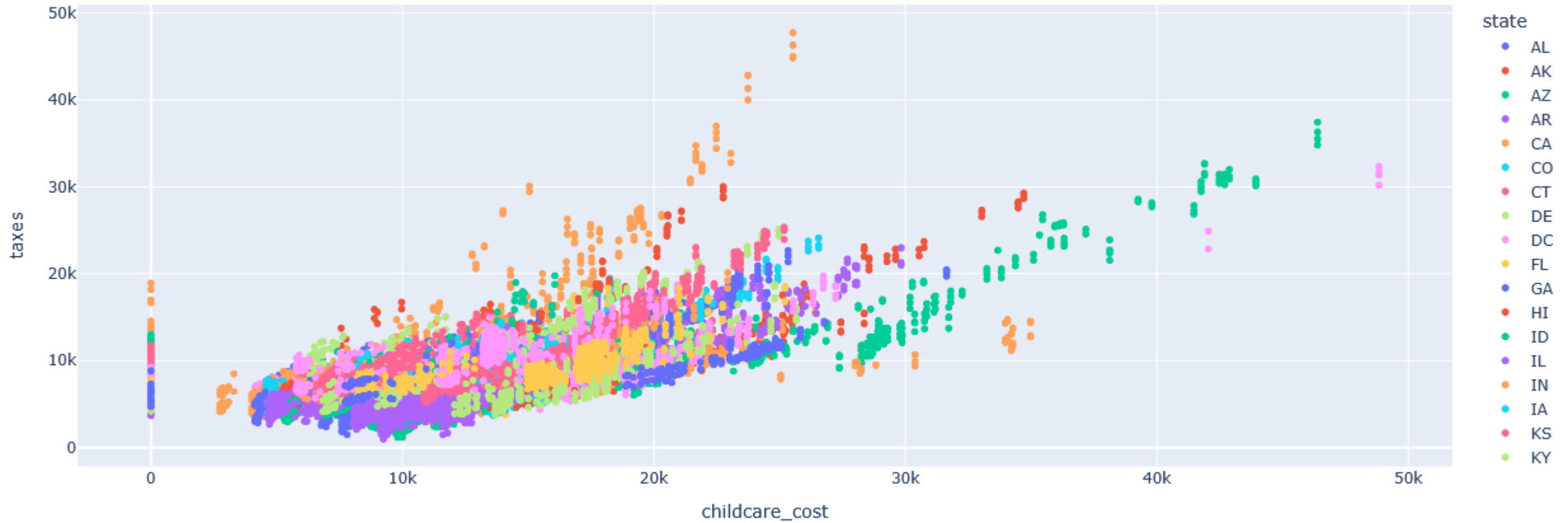
Total Cost vs. Transportation Cost by State

Total Cost vs. Transportation Cost by State



Childcare Cost vs. Taxes by State

Childcare Cost vs Taxes by State



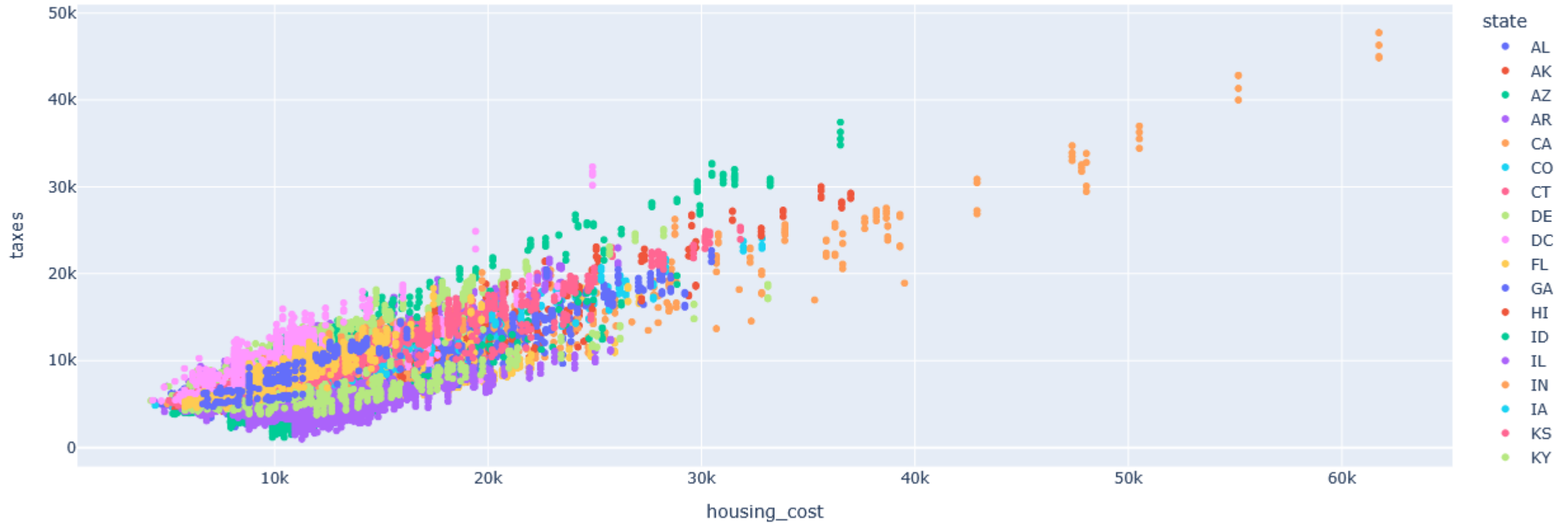
Median Family Income vs. Taxes by State

Median Family Income vs. Taxes by State



Housing Cost vs. Taxes by State

Housing Cost vs. Taxes by State



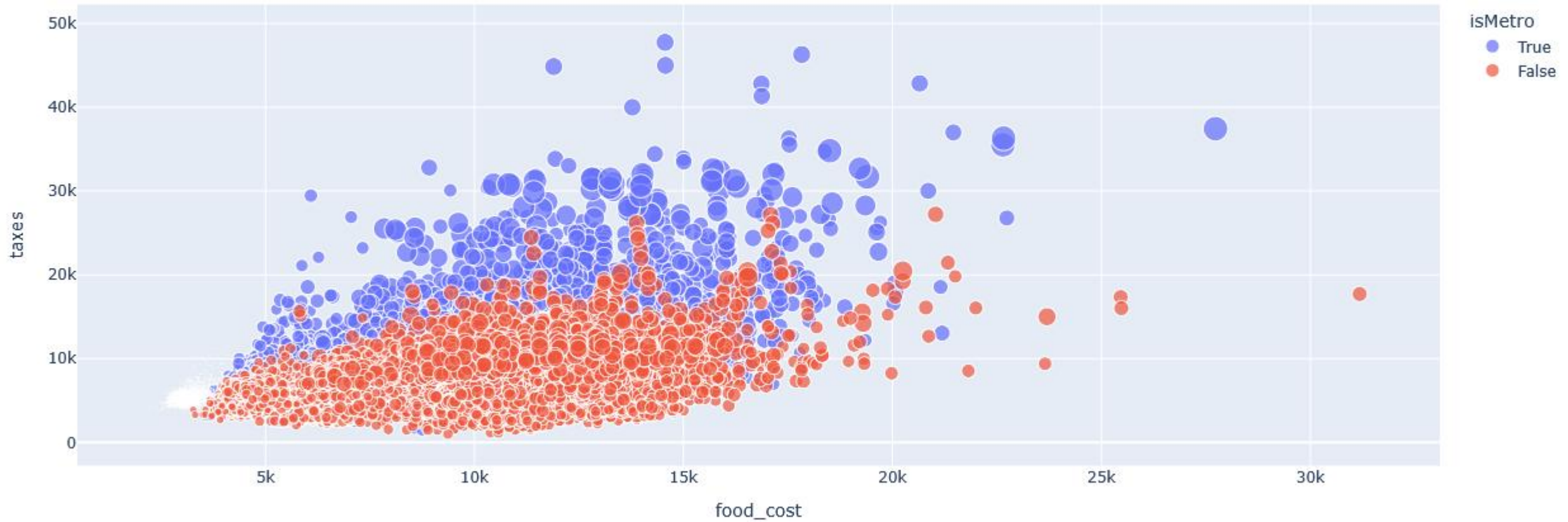
Housing Cost vs. Taxes Bubble

Housing Cost vs. Taxes sized by Transportation Cost

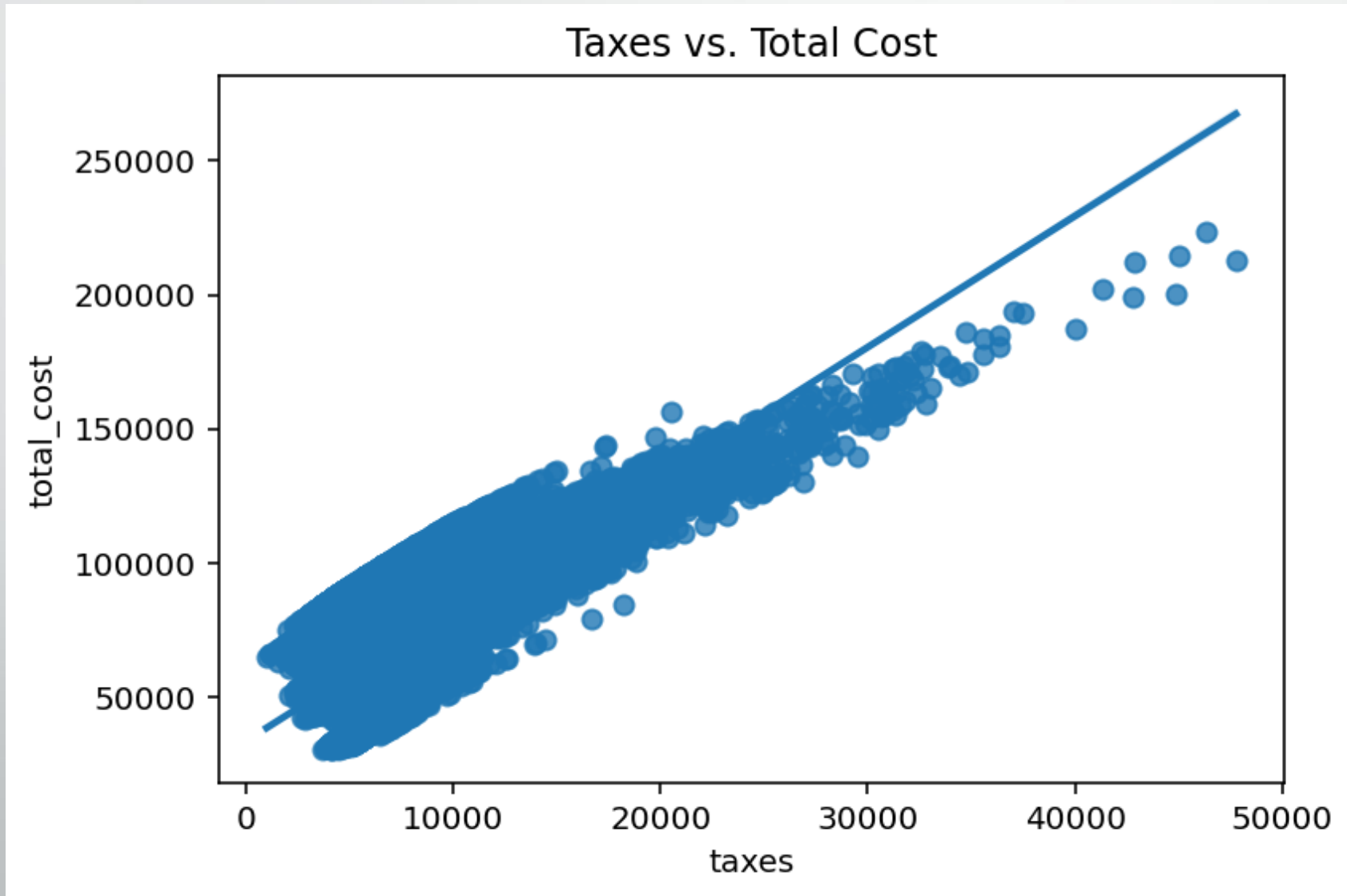


Food Cost vs. Taxes Bubble

Food Cost vs. Taxes sized by Childcare Cost

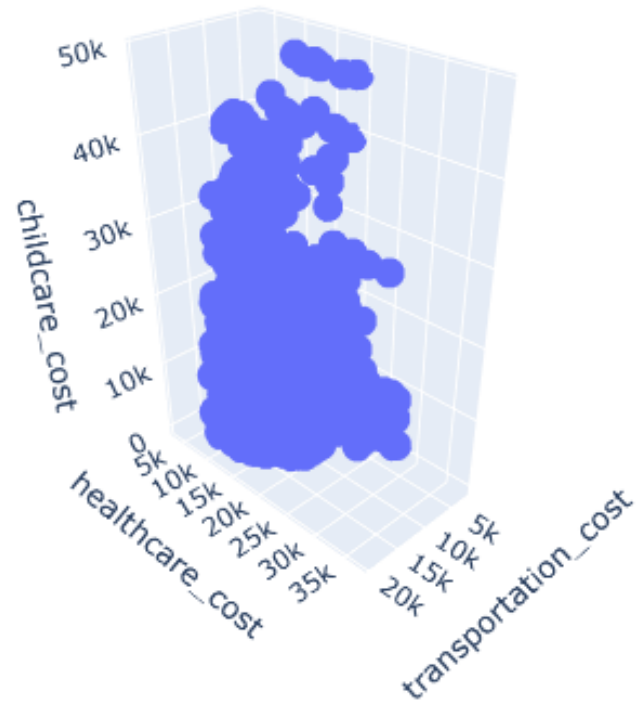


Taxes vs. Total Cost



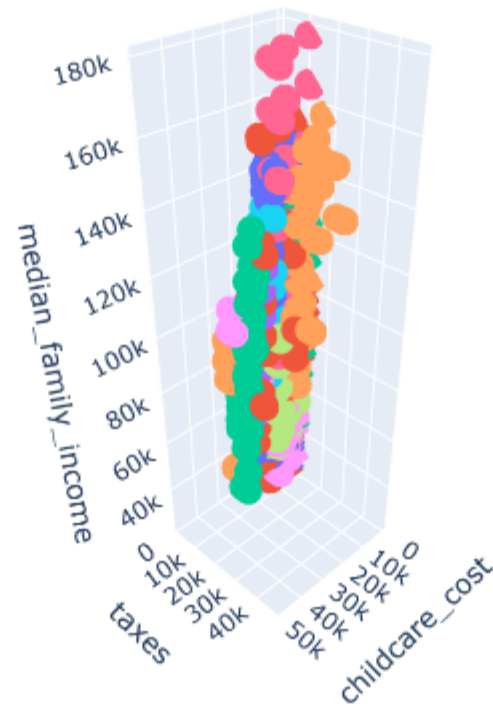
Transportation vs. Healthcare vs. Childcare

Transportation Cost vs. Healthcare Cost vs. Childcare Cost



Childcare vs. Taxes vs. Median Family Income

Childcare Cost vs. Taxes vs. Median Family Income by State

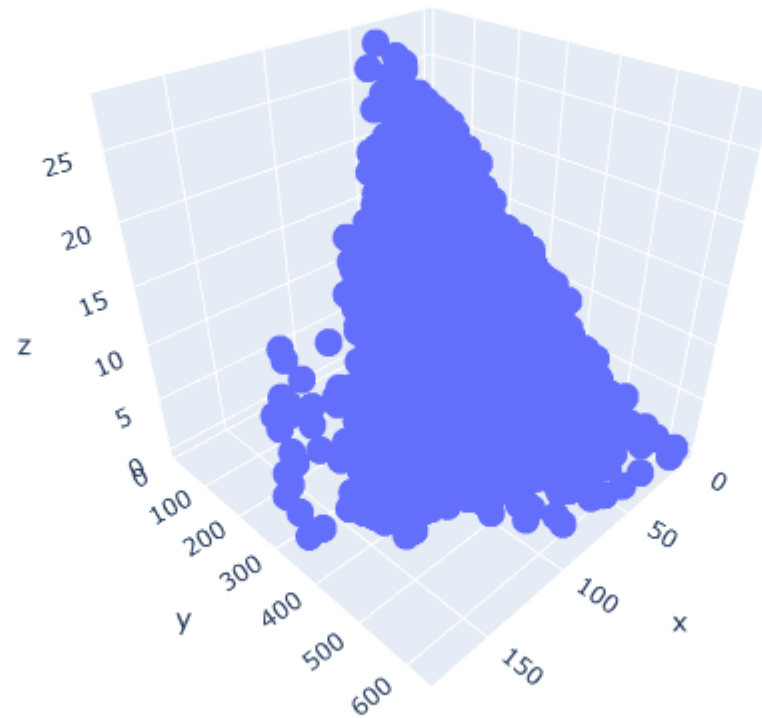


state

- AL
- AK
- AZ
- AR
- CA
- CO
- CT
- DE
- DC
- FL
- GA
- HI
- ID
- IL
- IN
- IA
- KS
- KY

3 Component NMF

3 Component NMF



Conclusions

- High correlations among costs
- Washington D.C. generally stands out
- Many means appear similar
- Non-Metro appears with low taxes